

Suggested Solutions to Exercises 3

1. (a) The risk-neutral probability is the probability of an up movement of the stock *in a risk-neutral world*. We know that this risk-neutral probability can be calculated by finding the probability that gives a return on the stock equal to the risk-free rate (in this risk-neutral world). Therefore the risk-neutral probability, $\hat{p}$ must satisfy

$$42\hat{p} + 38(1 - \hat{p}) = 40e^{0.08 \times 1/12}$$

so that $4\hat{p} = 40e^{0.08 \times 1/12} - 38$ which gives $\hat{p} = 0.5669$. The value of the option is then the expected discounted payoff under the risk-neutral probability, or

$$\begin{aligned} f &= [(42 - 39) \times \hat{p} + 0 \times (1 - \hat{p})] e^{-0.08 \times 1/12} \\ &= 3 \times 0.5669 \times e^{-0.08 \times 1/12} \\ &= \$1.69 \end{aligned}$$

Note that the risk-neutral probability can also be calculated as

$$\hat{p} = \frac{e^{rt} - d}{u - d}$$

where u and d are the up and down multipliers for the stock price. These are $u = 42/40 = 1.05$ and $d = 38/40 = 0.95$, so that

$$\hat{p} = (e^{0.08 \times 1/12} - 0.95)/(1.05 - 0.95) = 0.5669$$

- (b) We want to construct a portfolio consisting of an amount of the stock and cash that has the same payoff as the option at maturity. This must be worth 3 for a stock up movement, and 0 for a stock down movement. The portfolio will include, first, an amount of stock equal to

$$\frac{f_u - f_d}{S(u - d)} = \frac{3 - 0}{42 - 38} = \frac{3}{4}.$$

For the portfolio to be worth 0 in a down movement we must also include an amount of cash equal to

$$0 - \frac{3}{4} \times 38 = -28.5$$

at maturity. In an up movement the portfolio will then be worth

$$\frac{3}{4} \times 42 + 28.5 = 3$$

as required. By no arbitrage, the option will be worth the same as the portfolio today, which, after discounting the cash component to today, is

$$\frac{3}{4} \times 40 - e^{-0.08 \times 1/12} \times 28.5 = 1.69.$$

2. **Risk-neutral probability approach.** The probability of an up movement in a risk-neutral world, $\hat{p}$, must satisfy

$$55\hat{p} + 45(1 - \hat{p}) = 50e^{0.1 \times 0.5}$$

so that

$$10\hat{p} = 50e^{0.1 \times 0.5} - 45$$

which gives $\hat{p} = 0.7564$. Alternatively calculate $\hat{p}$ as

$$\hat{p} = \frac{e^{rt} - d}{u - d}$$

with $u = 55/50, d = 45/50$, which gives the same value. The value of the option, f say, is then its expected payoff under the risk-neutral probability, discounted at the risk-free rate, or

$$f = (0 \times 0.7564 + 5 \times 0.2436)e^{-0.1 \times 0.5} = 1.16.$$

3. We have a two-step binomial model for the stock price, so it is probably easier to use the risk-neutral probability approach rather than construct replicating portfolios. In this case $u = 1.10, d = 0.9$ and $r = 0.08$ so that

$$\hat{p} = \frac{e^{0.08 \times 0.5} - 0.9}{1.1 - 0.9} = 0.7041$$

The tree for stock price movements is shown. We can work back from the end of the tree to the beginning as indicated in the diagram to calculate the expected value of the option payoffs under the risk-neutral probabilities. This gives the value of the option as \$9.61, which can be calculated as

$$e^{-2 \times 0.08 \times 0.5} (0.7041^2 \times 21 + 2 \times 0.7041 \times 0.2959 \times 0 + 0.2959^2 \times 0) = 9.61$$

by discounting the expected payoffs under the risk-neutral probabilities over both time periods.

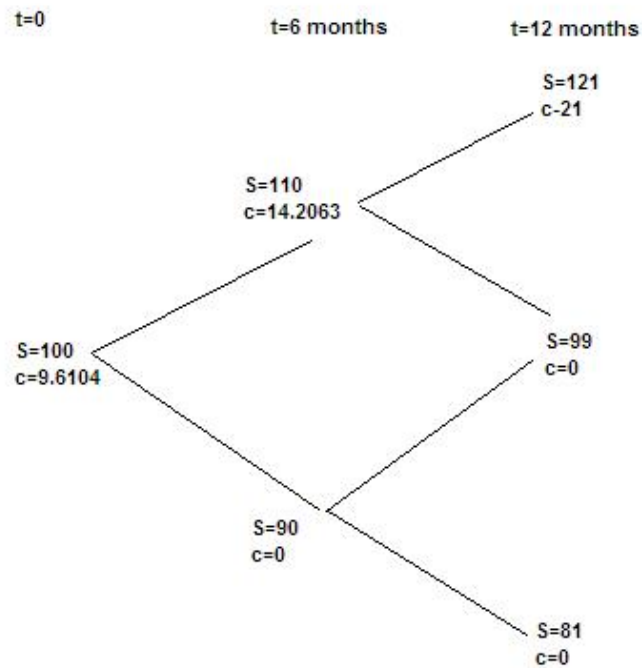


Figure 1: Tree for Q3

4. **Risk neutral approach.** Using risk-neutral probabilities we have

$$\begin{aligned}\hat{p} &= \frac{e^{rt} - d}{u - d} \\ &= \frac{e^{0.1 \times 2/12} - 0.92}{1.08 - 0.92} = 0.6050\end{aligned}$$

so that

$$f = e^{-0.1 \times 2/12} (0.6050 \times 729 + 0.3950 \times 529) = 639.3.$$

5. (a) The risk-neutral probability of a one-step up movement, $\hat{p}$, is

$$\begin{aligned}\hat{p} &= \frac{e^{rt} - d}{u - d} \\ &= \frac{e^{0.05 \times 3/12} - 0.95}{1.06 - 0.95} = 0.5689\end{aligned}$$

There is a payoff from the option of $56.18 - 51 = 5.18$ for the position after two stock up movements. For all other cases, the option will payoff zero. We can therefore calculate the value of the option as

$$f_{uu} \times \hat{p}^2 \times e^{-rt} = 5.18 \times 0.5689^2 \times e^{-0.05 \times 6/12} = 1.635$$

(b) The payoffs for the put option are

$$51 - 50 \times 0.95 \times 1.06 = 51 - 50.35 = 0.65$$

for the up/down (or down/up) movement, and

$$51 - 45.125 = 5.875$$

for two down stock movements. It is zero for two up stock movements, as the stock is above the strike price. The risk-neutral probabilities are the same as in part (a). Calculating the value of the option as the discounted expected payoff under the risk neutral probabilities gives

$$\begin{aligned}& (\hat{p}^2 f_{uu} + 2\hat{p}(1 - \hat{p})f_{ud} + f_{dd}(1 - \hat{p})^2) e^{-rt} \\ &= (2 \times 0.5689 \times 0.4311 \times 0.65 + 0.4311^2 \times 5.875) e^{-0.05 \times 6/12} \\ &= 1.376.\end{aligned}$$

This can be seen by working backwards through the relevant tree.

- (c) The put-call parity result says $P + S = C + Xe^{-rt}$. Here the value of the put plus the stock price is $1.376 + 50 = 51.376$, and the value of the call plus the present value of the strike price is $1.635 + 51e^{-0.05 \times 0.5} = 51.376$. This verifies that put-call parity holds.
- (d) To test whether it is worth exercising the option early we compare the value calculated for the option at each node in the tree with the payoff from immediate exercise. At the node for one down movement the payoff from immediate exercise is $51 - 47.5 = 3.5$. Since this is greater than the value of the option at this point, 2.8664, the option should be exercised early at this node. The option should not be exercised after one up movement, or today.
- (e) To calculate the value of the American put option we use the discounted expected payoff under the risk-neutral probabilities, using option values that are the maximum of the European option and the payoff for early exercise at each node. Using the result from part (d), this gives us a value of

$$\exp(-rt)\{\hat{p} \times 0.277 + (1 - \hat{p}) \times 3.5\} = 1.6458,$$

using $\hat{p} = 0.5689$ calculated previously. Note that this is greater than the 1.376 for the European call.

- (f) Early exercise of the call option after one stock up movement will pay $53 - 51 = 2$, and will pay zero after one down movement. By comparing values at the nodes of the tree diagram for the call option, it can be seen that this is less than or equal to the value of the European call option at each of these points. Therefore early exercise of the American option is never optimal in this case, and so the American call option is worth the same as the European call option, 1.635.

In fact this is an example of the more general result that it is never optimal to exercise an American call option early on a non-dividend paying stock. However, as we have seen in part (e), it may be optimal to exercise an American *put* option early.